BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI
Hyderabad Campus
SECOND SEMESTER 2025-2026

Mathematical Foundations for Data Science

Assignment - 1

Max Marks: 10 Last Date for submission: 7-March-2026

Instructions:

- Use Python coding to solve the following problems
- For each problem, the detailed python coding, the final and intermediate solutions page has to be attached
- Make a single PDF file and submit
- Write your name and ID number at the top of the first page
- Each question carries 2M

1. Find the Reduced Row-Echelon form of the matrix $\begin{bmatrix} 0 & 1 & 3 & -1 & 2 & 1 \\ 0 & -2 & -5 & 3 & 0 & 2 \\ 0 & 0 & 0 & 1 & -1 & 7 \\ 0 & 2 & 6 & -2 & 2 & 0 \\ 0 & 3 & 9 & -2 & 4 & 7 \end{bmatrix}$

2. Using Gauss Elimination, find the inverse of the matrix $\begin{bmatrix} 0 & 1 & -1 & 1 \\ 2 & 2 & 0 & -2 \\ 1 & 1 & -2 & 0 \\ 0 & 1 & 2 & 0 \end{bmatrix}$

3. Consider the set of vectors

$$\begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} -3 \\ 0 \\ 5 \end{bmatrix}$$

   Check whether this set of vectors a basis for $\mathbb{R}^3$.

4. Find the rank of the matrix A where $A = \begin{bmatrix} 1 & 2 & 0 & 1 & 1 \\ 2 & 4 & 1 & 3 & 0 \\ 3 & 5 & 2 & 4 & 1 \\ -4 & -3 & 1 & -2 & 1 \end{bmatrix}$

5. Compute the singular value decomposition (SVD) of the matrix $\begin{bmatrix} 3 & 1 & 1 & 0 \\ 1 & 2 & -2 & -1 \end{bmatrix}$